

Local-Perception-Enhanced Spatial–Temporal Evolving Graph Transformer Network: Citywide Demand Prediction of Taxi and Ride-Hailing

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Abstract—Accurate prediction of demand for traditional taxi and ride-hailing services is crucial for addressing supply-demand imbalances. However, recent studies based on global adaptive graphs, local spatial-temporal graphs, and self-attention mechanisms struggle to effectively capture the dynamic and intricate relations in demand. Moreover, existing dynamic graph generators face challenges in efficiently producing high-quality graphs to learn the diverse interactions among zones along time axis and their shared patterns spanning various time scales. To solve these challenges, we propose a novel Local-Perception-Enhanced Spatial-Temporal Evolving Graph Transformer Network (LPE-STGTN), aimed at improving the effectiveness and efficiency of extracting intricate local dependencies in taxi demand. Specifically, we elaborately design a spatial-temporal evolving graph generator to absorb shared and diversified inter-zone relations across different temporal periodicities and specific interactions among zones within each time step. Furthermore, an attention free transformer with local context (AFT-local) is introduced to effectively learn the correlations between adjacent time steps. Extensive experiments on three taxi datasets of New York and Beijing are carried out to evaluate the superior performance of our model. Compared with the most competitive baseline, our model achieves a balance between effectiveness and efficiency on three datasets, with average training time reduction of 70.66% and average performance improvement of 1.96%.

Index Terms—Traffic forecasting, dynamic graph generator, graph convolution network, transformer.

I. INTRODUCTION

RIDE-HAILING service and traditional taxi, collectively referred to as taxi, are two of the most common travel modes for commuting with the advantages of flexibility and comfort. While abundant excellent approaches for road traffic

flow prediction can serve as valuable references to deal with the difficulties in taxi demand, they still require refinement to adapt the characteristics of taxi demand.

Distinguished from taxi demand, traffic flow there is defined as the number of vehicles passing through a spatial unit, such as a road segment or traffic sensor point [1]. The difficulties in traffic flow prediction and taxi demand forecasting are similar but quite different. Generally, traffic flow and taxi demand both generate within a road network, and exhibit relatively stable spatial-temporal distributions from a global perspective while experiencing dynamic fluctuations at a local level [2].

Although the overall patterns are similar, the demand prediction of taxi encounters more serious challenges in capturing spatial-temporal dependences especially from a local perspective, compared with traffic flow forecasting. For the spatial dimension, traffic flow is essentially the number of vehicles passing through the observation nodes on a road network, naturally presenting strong continuity and prominent spatial relations in local area. As shown in Fig.1 (a), there is an explicit relationship among the traffic flow of neighboring nodes formulated as Eq. (1):

$$fl_1 = fl_2 = fl_3 + fl_4 + fl_5 \quad (1)$$

where fl_i represents the flow of node i . This explicit relationship in local area can be captured by the graph based on adjacency matrix, as shown in Fig.1 (c), while the correlations in traffic flow among distant nodes are too implicit to capture. By contrast, taxi demand is not just the number of orders, but also reflects the information of every true trip. The generation of taxi demand of one zone depends more on its own regional function and travel activities rather than neighboring zones. The relations in taxi demand of spatially separate zones are more evident than that in the traffic flow of distant nodes, due to the stable distribution of origin-destination (OD) caused by the long-term travel patterns [3]. As shown in Fig.1 (b) and (d), there are clear relationships represented as trips such as movements from residential areas to workplace or school between the left zones and right zones, which is formulated as Eq. (2) and Eq. (3):

$$q_{5,in} = q_{1,out} + q_{2,out} \quad (2)$$

$$q_{6,in} = q_{3,out} + q_{4,out} \quad (3)$$

where $q_{i,in}$ and $q_{i,out}$ represent the inflow and outflow of taxi demand in zone i . However, there is not more explicit

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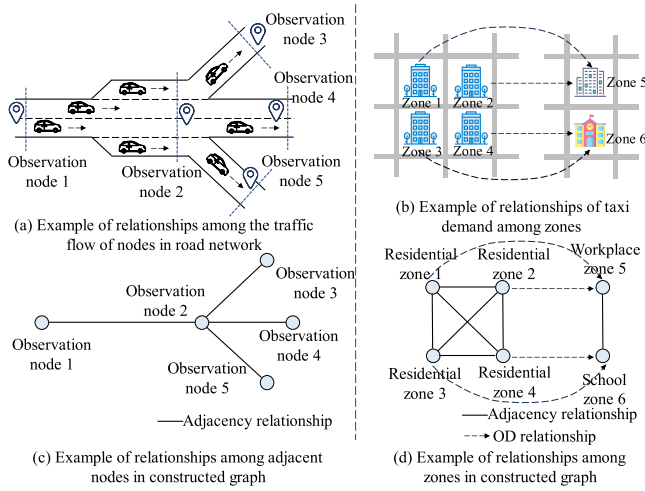


Fig. 1. Comparison between traffic flow and taxi demand.

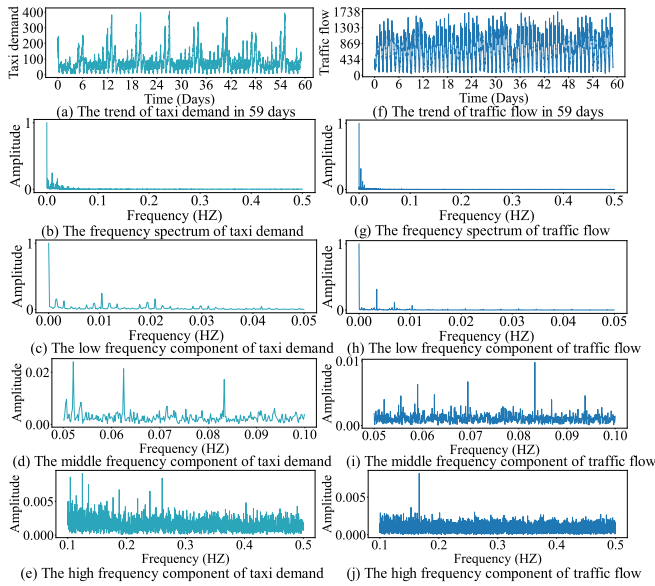


Fig. 2. Differences in local fluctuating patterns between traffic flow and taxi demand.

information in taxi demand of adjacent zones like the relationships represented as Eq.(1) in traffic flow of neighboring nodes. The graph in Fig.1 (d) is hard to mine more valuable interactions among neighboring zones in addition to adjacency relationship.

For the temporal dimension, both traffic flow and taxi demand exhibit obvious periodicity in long-term and dynamic variations in short-term, as illustrated by taxi demand in New York and traffic flow in California in Fig.2 (a) and (f). Even though the holistic temporal profiles are alike, there are crucial distinctions lying in the local trend between traffic flow and taxi demand. Due to the shared similarities with fluid flow and the connectivity of the road network, the local variation of traffic flow is continuous and gradual. Even if congestion or accidents happen, a long or short process is required for abnormal effects to propagate from upstream nodes to downstream nodes, rather than undergoing an instantaneous state transition [4]. In contrast, the local trend of

taxi demand presents pulse-like fluctuation. The high-intensity travel during peak periods or from transportation hub can result in a surge of trip requests in a split second. The differences in local fluctuating patterns between traffic flow and taxi demand are clearly reflected by the comparison between Fig.2 (a) and (f). Furthermore, Fast Fourier Transform (FFT) is used to mine the statistic differences between the two kinds of data in frequency domain. Subsequently, the frequency spectrum analysis across different frequency ranges is conducted, following the methods outlined in research [2]. The low-frequency components correspond to the periodicity in long-term, and high-frequency components align with local fluctuations. As shown in Fig.2 (b) and (g), the energy of taxi demand and traffic flow are both mainly concentrated in low-frequency, indicating that the two kinds of data exhibit strong periodic patterns. However, when it comes to high-frequency components in Fig.2 (e) and (j), the amplitude and oscillation range of taxi demand are more prominent than those of traffic flow, reflecting that the local dependencies of taxi demand are more dynamic and complex. In conclusion, the demand prediction of taxi is faced with more intricate challenges in capturing spatial-temporal dependences especially from a local perspective, compared with traffic flow forecasting.

Great efforts have been devoted to mining spatial-temporal dependencies in traffic data. Traditional statistical methods like autoregressive integrated moving average (ARIMA) and its variants [5], [6], [7] have been applied successfully for taxi demand prediction, but their performance is limited without considering spatial dependencies. Classic machine learning approaches [8], [9], [10] show improved performance but rely heavily on hand-designed features, failing to capture intricate spatial-temporal dependencies. Recent year has witnessed the development of traffic data mining techniques based on deep learning. Convolutional neural networks (CNNs) [11] and recurrent neural networks (RNNs) [12], [13] model spatial and temporal relations respectively, achieving impressive performance in Euclidean space. Graph neural networks (GNNs) [14], as the frontier of deep learning research, has shown state-of-the-art performance in modeling non-Euclidean space, which is more applicable for learning the underlying graph structure of traffic data.

While having demonstrated to be effective for traffic prediction, existing approaches still encounter limitations and challenges. First, there is an obvious limitation in recent studies with global-local view learning that they are inclined to separately identify global and local dependences from spatial or temporal view. Specifically, they tend to regard the global and local dependences as adjacent and non-adjacent relations in spatial dimension while extract holistic temporal correlations through gate recurrent unit (GRU) or 1D convolution [15], [16]. Alternatively, some studies capture global and local time information, meanwhile graph attention network (GAT) is used to learn global spatial relations [17], [18]. The above global-local learning view should be further improved to adapt the characteristics of traffic data which exhibits both complex spatial-temporal dependences in global and local perspectives. Second, the main approaches designed for local feature extraction of traffic flow struggle to effectively extract more

dynamic and intricate relations in taxi demand. On the one hand, road network-based adjacency matrix plays an important role in constructing localized graph [19] and add local information into transformer [20] for the extraction of local spatial-temporal dependencies. However, it may not be the best matrix to represent the inter-zone relations for taxi demand, because there is no explicit relationship such as Eq. (1) like traffic flow of neighboring nodes among the taxi demand of neighboring zones. On the other hand, researches based on global adaptive graph [21], [22] or attention mechanism [23] learn the overall spatial interaction among zones adaptively, but are hard to extract the time-varying spatial-temporal dependencies. Third, current generators of time-varying graphs are inefficient with complex design, and lacks sufficient representation of associations among different time periods and zones. The existing generators of time-varying graphs typically process each time-step's raw input through a series of complex transformations, such as multiple linear transformations [24], spatial self-attention mechanisms [4], [25], fusion with hidden states from GRU [26], [27], which is further processed by GCN and learnable parameter matrix [28]. It should be noted that the above generators of dynamic graphs encounter challenges in generating high-quality graphs that effectively capture the diverse interactions among zones over time and their shared patterns spanning various time scales. Moreover, dynamic graph generators based on multi-head self-attention mechanism or hidden states from GRU both suffers from the expensive computation costs due to the inherent limitation of transformer and GRU in efficiency.

Motivated by the above limitations and challenges, we propose a novel Local-Perception-Enhanced Spatial-Temporal Evolving Graph Transformer Network (LPE-STGTN) to improve the effectiveness and efficiency of extracting intricate local dynamics in taxi demand. For this goal, we design a local spatial-temporal intertwined module consisting of attention free transformer with local context (AFT-local) and a spatial-temporal evolving graph generator. Specifically, we introduce the AFT-local [29] into our model to efficiently characterize the dynamic correlations among inter-node relationships across adjacent time steps. Moreover, a spatial-temporal evolving graph generator is designed for describing the dynamic interaction among zones along time axis, where the multi-scale embedding matrices are designed to efficiently absorb shared and diversified inter-node relations across different temporal periodicities, meanwhile GAT is used to self-learn the specific spatial interaction among nodes within each time step. In addition, global spatial-temporal dependencies are also captured by global spatial-temporal regularity module based on dual semantic GCN and transformer. The global-local multi-grained features are further fused by global-local information fusion module consisting of a multi-head self-attention mechanism and a single-layer convolution. Through these effective and efficient designs, our model is competent in capturing local intricate spatial-temporal dependencies and global spatial-temporal regularity in taxi demand, thereby gaining considerable performance and efficiency improvement compared with current GCN-based methods in extensive experiments.

II. LITERATURE REVIEW

A. Deep Learning in Traffic Forecasting

For traffic forecasting, the fundamental challenge is to effectively extract the spatial-temporal dependencies in traffic data. Generally, CNN is employed to extract the spatial dependencies while RNN and its variants are utilized to learn temporal correlations. Many current studies follow such paradigms and explore different combinations of the two kinds of networks [30], [31], [32], but CNN-based approaches are not optimal for spatial-temporal traffic data with graph-based forms. Recent studies have shed light on the promising ability of GCN to adapt to the graph structure hidden in traffic data. STGCN [33] treats traffic data as graph-structured time series and replaces recurrent units with 1D convolutional layers. Another strategy is to combine GCN with attention mechanism, such as ASTGCN [34] and GMAN [35]. In addition, some researches [36], [37], [38] extend the single-graph convolution to multi-graph for extracting kinds of explicit dependencies. However, the above studies depend on fixed graph structure which tends to learn the kinds of explicit relations but ignores the abundant implicit information. The subsequent works such as Graph WaveNet [22], AGCRN [21] and CCRNN [39] notice the limitation of the above methods and introduce the adaptive graph based on parameterized adjacency matrix to self-learn the relations between nodes. Although the combination of fixed and adaptive graphs is contributed to learn the explicit and implicit information at a holistic level, these studies struggle to hierarchically absorb such information across multiple scales. In other words, they fail to deeply capture the global and local spatial-temporal heterogeneity in traffic data, featured as stable spatial-temporal distributions from a global perspective and dynamic fluctuations at a local level.

B. Global-Local Learning in Traffic Forecasting

To enhance the ability of hierarchical extraction, recent studies start to learn spatial-temporal dependencies with global-local view learning, where modules are specifically designed to mine multi-level relations from global and local perspectives. Several studies [15], [16], [40] tend to define the global and local dependencies as adjacent and non-adjacent relations in spatial dimension, where global static and local adaptive adjacency matrices are designed to capture global and local spatial dependencies, while temporal modules based on GRU or 1D convolution are employed to learn holistic time correlations. In contrast, GLGAT [17] and Wang et.al. [18] focus on grasping global and local time information by self-attention and GRU respectively, meanwhile GAT is applied to learn global spatial relations. There is an obvious limitation that the above studies are inclined to separately identify global and local dependencies from spatial or temporal view. Actually, traffic data typically exhibit a stable spatial-temporal distribution in a global perspective but demonstrate spatial-temporal dynamics and uncertainty in a local level. Therefore, coupling consideration of spatial-temporal dependencies from global and local perspectives is more suited to the characteristics of traffic data. STSGCN [19] proposes a novel manner to

construct a localized spatial-temporal graph based on matrices extension for learning spatial dependencies within three timesteps, consisting of temporal connectivity graph and traditional spatial graph. Fang et.al. [20] propose Lastjormer to learn spatial-temporal dependencies from global and local perspectives, based on a spatial-temporal joint attention. Although the above approaches extract global and local dependencies by coupling consideration of spatial-temporal dimension, whether the localized spatial-temporal graph or multi-hop diffusion convolution in spatial-temporal joint attention is both constructed based on distance adjacent matrix, which is more suited to traffic flow prediction rather than taxi demand forecasting. Moreover, when faced with more dynamic and complex dependencies hidden in taxi demand, current local components designed for traffic flow may be struggle to effectively extract dynamic and implicit relations for superior forecasting performance.

C. Local Dependencies Extraction in Traffic Forecasting

Local dependencies, as described earlier, are coupling consideration of spatial-temporal dependencies instead of separate extraction. Specifically, local dependencies include the short-term fluctuation and uncertainty in temporal dimension and resulting dynamic spatial interaction among zones. GRU combined with global adaptive graphs is a classical way to extract dynamic spatial-temporal information [18], [23], [41], where the adaptive graphs tend to learn overall spatial dependencies from all the data sequences. Meanwhile, high-quality extraction of short-term relations excessively depends on the depth of GRU requiring high computation cost. Another strategy is to enhance the local awareness of transformer [20], which essentially learns local dependencies from global representation embedded with local information rather than directly extracting true and pure local dependencies. Additionally, a promising way to extract such dependencies is design a dynamic graph generator to obtain dynamic graph. According to the number of timestamps fed into the dynamic graph generator, current studies that focus on dynamic graph can be broadly categorized into two groups. The first category is sequence-varying graph, such as [28] and [42], where the graph generator designs a dynamic graph for every sequence involved multiple time steps. As the traffic sequence becomes longer, the interaction among zones generated from the whole sequence is more likely to reveal commoner patterns rather than reflecting the time-varying and heterogeneous spatial-temporal information. By contrast, generating time-varying graph for every time step is a more effective method to learn dynamic dependencies. PDFormer [4] and Zheng et al. [25] feed the input of every timestamp into self-attention to generate time-varying graph, overlooking the evolving information between adjacent time steps. Li et.al. [27] concatenate the node information of every timestamp and hidden states of former timestamps as the input of multiple graph convolution layer to generate time-varying graph. Similarly, Xia et.al. [26] integrate the hidden state of former time step and the current input transformed by multi-head attention mechanism to construct dynamic representation between nodes. However, the above generators of dynamic graphs are hard to produce

high-quality graphs to learn the diverse interactions among zones along time axis and their shared patterns spanning various time scales. Moreover, dynamic graph generators based on multi-head self-attention mechanism or hidden states from GRU both suffer from the expensive computation costs due to the inherent limitation of transformer and GRU in efficiency.

D. Research Gaps and Contributions

In conclusion, existing studies focused on taxi demand prediction still face the following limitations and challenges:

- 1) **Limited ability to extract coupling spatial-temporal dependencies from global and local perspectives.** Current studies are inclined to separately identify global and local dependences from spatial or temporal view, which should be further improved to adapt the characteristics of traffic data with both complex spatial-temporal dependences in global and local perspectives.
- 2) **Challenge in capturing more dynamic and intricate relations.** The current main approaches designed for local extraction of traffic flow, such as global adaptive graphs, local spatial-temporal graphs, self-attention mechanisms and transformer with local information, struggle to effectively extract the more dynamic and implicit relations hidden in taxi demand.
- 3) **Inefficient and complex design of time-varying graphs.** The generators of dynamic graphs in existing studies are hard to produce high-quality graphs to learn the diverse interactions among zones along time axis and their shared patterns spanning various time scales, meanwhile the complex design brings the expensive computation costs.

To handle these drawbacks, we propose a novel LPE-STGTN for taxi demand prediction. The main contributions of this work are as follow:

- 1) We propose a novel LPE-STGTN, which aims to improve the effectiveness and efficiency by considering the spatial-temporal coupling dependencies from global and local perspectives.
- 2) Different from existing models limited by insufficient ability to extract spatial-temporal coupling dependencies from a local perspective, a local spatial-temporal intertwined module, based on an AFT-local and a spatial-temporal evolving graph generator, is designed to effectively extract all the dynamic dependencies in taxi demand.
- 3) In contrast to current methods for generating time-varying graphs with inefficient and complex designs, a spatial-temporal evolving graph generator is proposed, where multi-scale embedding matrices are designed to absorb shared and diversified inter-node relations cross different temporal periodicities, meanwhile GAT is used to self-learn the specific spatial interaction among nodes within each time step.
- 4) Extensive experiments are conducted on three real-world taxi datasets from New York and Beijing. Compared with the most competitive baseline, our model achieves

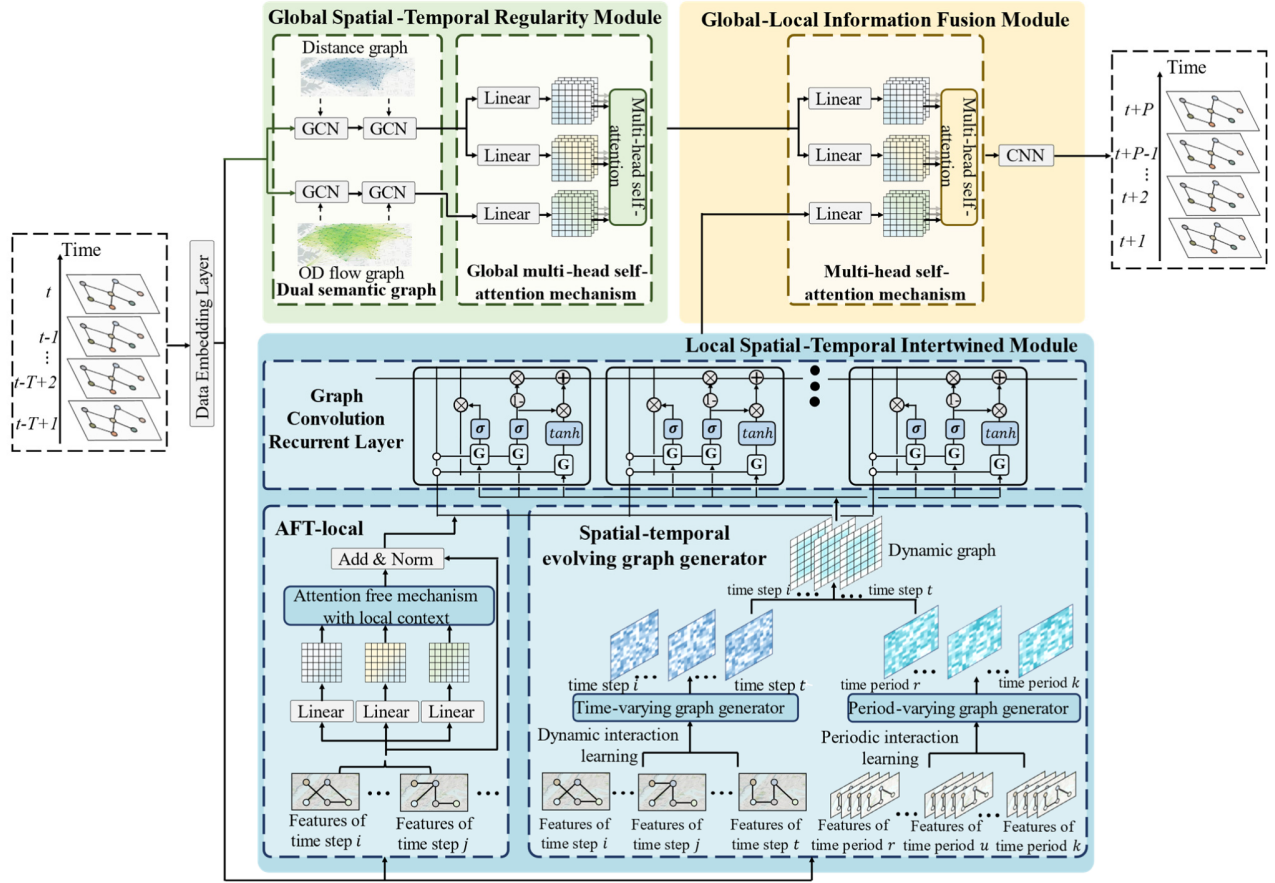


Fig. 3. Architecture of LPE-STGTN model.

a balance between effectiveness and efficiency on three datasets, with average training time reduction of 70.66% and average performance improvement of 1.96%.

III. METHODOLOGY

A. Problem Formulation

We first give several fundamental definitions to formalize the taxi demand forecasting problem.

Graph: The research area is first partitioned into many taxi zones which act as the basic spatial units in the following analysis. Let graph $G=(V, E, A)$ be the constructed graph, where $V = \{v_1, v_2, \dots, v_i, \dots, v_N\}$ represents a set of taxi zones, E is a set of edges connecting taxi zones, and N is the number of taxi zones. The correlations among the zones are formulated as an adjacency matrix $A \in R^{N \times N}$. Various relations reflected by different adjacency matrices are capable of constructing kinds of graph to capture abundant hidden information.

Taxi demand: The total time period is portioned to multiple time steps with a constant interval such as 5min or 15min, etc. The number of taxi orders generating in t -th time step at zone i is aggregated as taxi demand represented as x_t^i . Taxi demand of all the taxi zones in t -th time step is denoted as $X_t = \{x_t^1, x_t^2, \dots, x_t^N\}$.

Multistep taxi demand prediction: The purpose of taxi demand prediction is to learn a mapping function $f(\cdot)$ for

predicting the taxi demand at next P time steps based on known demand data at the first T time steps and constructed graph G . This is defined as follow:

$$[X_{t+1}, \dots, X_{t+P}] = f([X_{t-T+1}, \dots, X_t]; G) \quad (4)$$

where $P(P > 1)$ is the predictive time length and $T(T > 1)$ is the number of history time steps.

B. Overview Architecture

Fig.3 illustrates the framework of our proposed LPE-STGTN, consisting of a data embedding layer, a local spatial-temporal intertwined module, a global spatial-temporal regularity module and a global-local information fusion module. Specifically, a taxi demand sequence and corresponding temporal periodical information are concatenated and fed into the data embedding layer which converts the input data into high-dimensional space through convolution. Then, the output is processed by local spatial-temporal intertwined module, which is the core module to extract all the local dynamic dependencies in taxi demand. Meanwhile, global regularity module is also designed to capture and dynamically integrate kinds of semantic information from the output of data embedding layer. Both the fusion of local and global spatial-temporal information, and the completion of downstream task of taxi demand prediction are finished through a global-local information fusion module.

C. Data Embedding Layer

To model the periodicity hidden in taxi demand and enrich input features, we explicitly inject temporal periodical information into taxi demand sequence by encoding time-of-day index and day-of-week index following previous works [4], [35], [47]. Specifically, the time index t of X_t is transformed into day-of-week index $X_{w,t} \in R^7$ and time-of-day index $X_{d,t} \in R^1$, respectively. X_t , $X_{w,t}$ and $X_{d,t}$ are concatenated and fed into 1D convolution and obtain the embedding vectors as:

$$X_{t,emb} = Conv([X_t, X_{d,t}, X_{w,t}]) \quad (5)$$

$X_{t,emb}$ will be fed into the following local and global modules, and we still use X_t to replace $X_{t,emb}$ for convenience.

D. Local Spatial-Temporal Intertwined Module

1) Spatial-Temporal Evolving Graph Generator:

a) *Generation of time-varying graph:* Most recent studies apply GCN to traffic forecasting following the calculations designed in the spectral domain [43], [44], which is further generalized to high-dimensional form as follow:

$$GCN(A, X) = (I_N + D^{-\frac{1}{2}}AD^{-\frac{1}{2}})XW \quad (6)$$

where $I_N \in R^{N \times N}$ is an identity matrix, $A \in R^{N \times N}$ is the adjacency matrix of the graph, $D \in R^{N \times N}$ is the degree matrix and $D_{ii} = \sum_j A_{ij}$, $X \in R^{N \times C}$ is the input of the GCN layer, and $W \in R^{C \times F}$ is the learnable parameter, respectively. C and F are the input dimension and output dimension. However, Eq. (6) has an obvious limitation that it implies no matter which time period X belongs to, the adjacency matrix A is fixed. When applying to taxi demand forecasting, fixed adjacency matrix can only capture the static spatial relations in total time period, which fails to perceive the intricate fluctuation and uncertainty in diversified demand patterns during different time steps in reality.

Intuitively, it is necessary to construct time-varying graph to learn evolving spatial relations among zones along time axis. In other words, the taxi demand of every time step is supposed to assign an adjacency matrix to capture the specific spatial interaction between zones. Nevertheless, there is not enough prior information to design appropriate adjacency matrix for every time step. Although some studies [21], [22] has attempted to design a global adaptive adjacency matrix driving by original data completely for all the demand sequences, it is infeasible to apply these methods directly to generate adaptive adjacency matrices for each time step. For example, in terms of M time steps, assigning specific adjacency matrix for each time step will cause the scale of A becomes $R^{M \times N \times N}$, which is too huge to learn and would bring about overfitting problem especially when M is big.

Therefore, GAT is introduced to generate unique time-varying graph according to the features of each time step so as to self-learning the diversified spatial interaction among nodes along time axis, which is a more efficient way than parameterized matrix learning. Specifically, X_t is converted to h_t , and then combined with a weight vector α to calculate

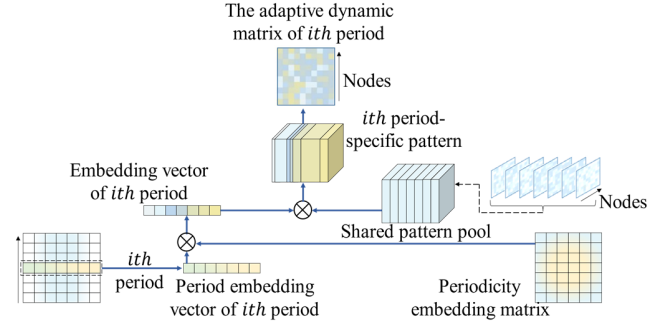


Fig. 4. The schematic diagram of multi-scale embedding matrices.

a pairwise attention coefficient $e_t^{i,j}$ between zone i and j . In order to stabilize the learning process, a dropout layer is added before the multiplication between e_t and h_t . The generation of unique time-varying graph can be formulated as follows:

$$h_t = X_t \cdot W \quad (7)$$

$$e_t^{i,j} = \text{Softmax} \left(\text{LeakyReLU} \left(\left[h_t^i, h_t^j \right] \cdot \alpha \right) \right) \quad (8)$$

$$A_t^{time} = \text{Softmax} (\text{Dropout} (e_t) \cdot h_t) \quad (9)$$

where X_t is the demand of all the zones within t -th time step, $W \in R^{C \times N}$ is the learnable parameter and $A_t^{time} \in R^{N \times N}$ is the unique time-varying graph of t -th time step, $[\cdot]$ represents the operation of concatenation.

b) *Generation of period-varying graph:* However, the specific time-varying graphs can only learn the heterogeneous spatial relations within different time steps, failing to reveal varieties of shared interaction patterns cross different time periods. Actually, each time period possesses its own unique or main travel activities due to the routine of daily life, such as the commuting movements from home to workplaces on weekday mornings and reverse direction in weekday evenings. The interactions between nodes within different time steps in the same time period exhibit not only time-varying diversities but period-specific commonalities.

Following this insight, multi-scale embedding matrices component are designed to absorb shared and diversified inter-node relations cross different intervals and periodicities, as shown in Fig.4. To alleviate the computational cost, the component only learns three smaller parameter matrices instead of directly assigning each time period with a parameterized adaptive matrix, which draws insights from matrix factorization and the periodicity of taxi demand. Specifically, the three smaller parameter matrices are a time period embedding matrix $E_p \in R^{h \times w}$ where h is the number of time periods and w is the periodicity dimension, a periodicity embedding matrix $E_d \in R^{w \times d}$ where d is the embedding dimension, and a shared pattern pool $W_s \in R^{d \times N \times N}$, respectively.

To enhance interpretability and reflect semantic information, we take h and w as 24 and 7 based on the time-series characteristics of taxi demand, indicating that 24 hours in a day are corresponding to 24 time periods and 7 days in a week reflects weekly periodicity. In other words, the time period embedding matrix E_p consists of 24 vectors, where each

vector learns the specific information of a corresponding hour and correlation between different days of a week. Similarly, the periodicity embedding matrix E_d is designed to adaptively learn the week periodicity in traffic demand. The shared pattern pool W_s contains a series of candidate interaction patterns denoted as adjacency matrices, which are the basis to form a new node interaction pattern according to a weight vector calculated by the multiplication. Then, $A \in R^{h \times N \times N}$ can be generated by $A = E_p \cdot E_d \cdot W_s$, consisting of h period-specific adjacency matrices.

From the view of i -th period, the period embedding vector of i -th period is selected from E_p and multiplied by E_d , where the result is expressed as the embedding vector of period i fusing the characteristics of i -th period and weekly periodicity comprehensively. Period-varying adjacency matrix A_i^{period} is extracted from W_s according to the embedding vector of period i , which can be interpreted as the process of learning unique interactions among nodes in period i from several alternative patterns discovered from all the traffic demand sequences. Finally, according to X_t during time period i , the generation of period-varying graph can be formulated as follows:

$$A_i^{period} = E_p^i \cdot E_d \cdot W_s \quad (10)$$

$$\tilde{A}_i^{period} = I_N + D^{-\frac{1}{2}} A_i^{period} D^{-\frac{1}{2}} \quad (11)$$

GAT and multi-scale embedding matrices component consist spatial-temporal evolving graph generator. The final spatial-temporal evolving graph A_t^{evlv} is the integration of time-varying graph and period-varying graph, where $LN()$ represents layer normalization.

$$A_t^{evlv} = \text{Softmax} \left(LN \left(A_t^{time} + \tilde{A}_i^{period} \right) \right) \quad (12)$$

2) *Attention Free Transformer With Local Context*: The dynamic graphs generated from the spatial-temporal evolving graph generator, integrate the spatial-temporal coupling heterogeneity across different time steps and periods. However, they struggle to capture the inter-information passing between adjacent time steps. Existing studies [18], [23], [41] often employ GRU layers to capture such local dynamics, leveraging their capability to propagate hidden states from previous time steps to subsequent ones. However, the quality of information in the hidden states excessively relies on the depth of GRU layers, which is effective but inefficient.

AFT-local is introduced to effectively and efficiently learn the information passing between adjacent time steps. Specifically, the matrix multiplication among key K_l , value V_l and query Q_l in transformer is replaced with element-wise multiplication. Moreover, a set of learned position biases $w \in s \times s$ with local window is multiplied with key to focus on the local context in adjacent time steps, where the local window is employed to mask tokens outside the window as 0, thereby preserving only the local information within the coverage of window. Specially, we add layer normalization before the exponential operation of key to stabilize the learning process. AFT-local can be formulated as follows:

$$Q_l = X_T W_l^q, K_l = X_T W_l^k, V_l = X_T W_l^v \quad (13)$$

$$X_t^l = \sigma \left(Q_{t,l} \right) \odot \frac{\sum_{t'=1}^T \exp(w_{t,t'}) \odot \exp(LN(K_{t',l}) \odot V_{t',l})}{\sum_{t'=1}^T \exp(w_{t,t'}) \odot \exp(LN(K_{t',l}))} \quad (14)$$

$$w_{t,t'} = \begin{cases} w_{t,t'}, & \text{for } |t - t'| < s \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

where W_l^q, W_l^k, W_l^v are learnable parameters; X_t^l is the local representation with local spatial-temporal context. Compared with GRU-based approaches, the advantages of AFT-local in capturing local information between adjacent time steps are significant, where the local view offers flexibility in focusing on a wider or narrower local context, and the calculation of inter-information between adjacent time steps is parallel rather than iterative.

3) *Graph Convolution Recurrent Layer*: To further mine the local dynamics and fluctuation in taxi demand, the output of AFT-local is fed into a graph convolution recurrent layer consisting of GRU and spatial-temporal evolving graph generator, where the diversified interaction within each time step, inter-information between adjacent time steps and shared patterns in different temporal periods are coupling learned and considered. Specifically, GCN based on spatial-temporal evolving graph replace the multi-layer perception layers in GRU. Moreover, only one GRU layer is required to coupling learn the multi-scale dynamic information, due to the strong ability in local perception of AFT-local and spatial-temporal evolving graph. The graph convolution recurrent layer is defined as follows:

$$r_t = \sigma \left(A_t^{evlv} \left[X_t^l, H_{t-1} \right] W_r + b_r \right) \quad (16)$$

$$u_t = \sigma \left(A_t^{evlv} \left[X_t^l, H_{t-1} \right] W_u + b_u \right) \quad (17)$$

$$C_t = \tanh \left(A_t^{evlv} \left[X_t^l, r_t \odot H_{t-1} \right] W_C + b_C \right) \quad (18)$$

$$H_t = u_t \odot H_{t-1} + (1 - u_t) \odot C_t \quad (19)$$

where H_t denotes the hidden state at time step t , $\odot$ is the Hadamard product, σ and $\tanh$ are the activation functions, $[\cdot]$ represents the operation of concatenation, the W_r, W_u, W_C, b_r, b_u and b_C are learnable parameters. Multi-scale dynamic information is fully learned in the graph convolution recurrent layer and alleviate the insufficient extraction of coupling spatial-temporal dependences caused by post-fusion efficiency [41]. The final output of local dynamic module is denoted as H_L .

E. Global Spatial-Temporal Regularity Module

1) *Dual Semantic Graph Convolution*: Compared with spatial-temporal intertwined module focused more on extracting the dynamic patterns of taxi demand in the local perspective, global regularity module is designed to capture the global patterns reflected in the regularity of commuting and the OD distribution, where the learned dependencies tend to be stable to characterize the global state of taxi demand. Thus, capturing the inherent dependencies highly related to taxi demand is the key to uncovering the regularity of taxi demand in the global perspective. Dual semantic graph convolution is proposed to capture the key semantic information which

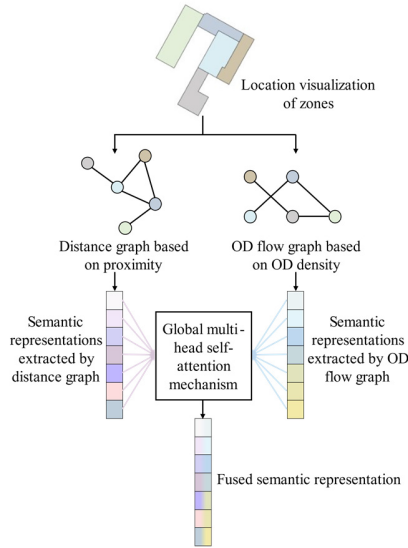


Fig. 5. The schematic diagram of global regularity module.

tends to reflect the global state of all the zones and acts as the supplement of local information. Considering that the overall state of taxi demand is closely correlated with demand density and proximity among zones, dual semantic graph convolution models two types of correlations among zones with graph, including distance graph and OD flow graph. The fusion of distance and OD flow graphs is contributed to learn the external influence from spatial proximity of zones, meanwhile the intrinsic influence from OD distribution and travel interactions among zones on taxi demand. The structure of global spatial-temporal regularity module is shown in Fig.5.

Distance graph is designed to measure the spatial proximity among zones in real traffic network. The element values in distance adjacency matrix determine the association strength between zones, indicating that the closer the two zones to each other, the more similar traffic demand patterns are shared by two neighboring zones. The distance adjacency matrix A_{dis} is defined as following:

$$a_{ij,dis} = \begin{cases} \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right), & i \neq j \text{ and } \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right) \geq \varepsilon_{dis} \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

where $a_{ij,dis}$ is the element in A_{dis} , d_{ij} is the spatial distance between zone i and j , σ^2 and ε_{dis} are the threshold to control the distribution and sparsity of A_{dis} . According to distance graph, the graph convolution based on A_{dis} is defined as follow:

$$GCN(A_{dis}, X) = \widetilde{A_{dis}} X W_{dis} \quad (21)$$

where $\widetilde{A_{dis}}$ is the normalized adjacency matrix with self-loop of distance graph and W_{dis} is the learnable parameters of distance graph.

Though distance graph is capable of portraying the spatial relationships among zones, it only focuses on the information aggregation for close zones without considering the situation

that some distant zones still have similar and related traffic demand pattern. For example, most residential zones are faced with high order demand in the morning and low travel requests in the evening on weekdays, while the situations are always contrary to distant working zones. From the semantic perspective of travel behavior, these correlations can be modeled by OD flow between zones naturally. In OD graph, although two zones are not close in spatial view, they become neighbors and exchange information when the OD flow between the two zones is large enough. The OD flow adjacency matrix A_{OD} is formulated as following:

$$a_{ij,OD} = \begin{cases} \left(\frac{flow_{ij}}{flow_{std}}\right)^2, & i \neq j \text{ and } \left(\frac{flow_{ij}}{flow_{std}}\right)^2 \geq \varepsilon_{OD} \\ 0, & \text{otherwise} \end{cases} \quad (22)$$

where $a_{ij,OD}$ is the element in A_{OD} , $flow_{ij}$ is the OD demand between zone i and j , $flow_{std}$ is the standard deviation of flow calculated by all the OD pairs, ε_{OD} are the threshold to control the distribution and sparsity of A_{OD} . Based on OD graph, the graph convolution with A_{OD} is formulated as follow:

$$GCN(A_{OD}, X) = \widetilde{A_{OD}} X W_{OD} \quad (23)$$

where $\widetilde{A_{OD}}$ is the normalized adjacency matrix with self-loop of OD flow graph and W_{OD} is the learnable parameters of OD flow graph.

Dual semantic graph convolution can be defined based on the distance graph and OD graph, not only capturing the spatial relationships but also modeling the travel interactions among zones. It is crucial to integrate them into a single module for information completeness and the integration mechanism is introduced in the next section.

2) *Global Multi-Head Self-Attention Mechanism*: Conventional multi-graph modeling usually adopts summation or concatenation to aggregate static relations extracted by adjacency matrix, which undermines its global learning ability. Due to the competitive performance of transformer in modeling global dependencies, a global multi-head self-attention mechanism is designed to enhance the global learning ability of the proposed dual graph convolution as shown in Fig.5, which fuses multiple semantic information adaptively according to their importance concerning model performance and enhances the critical information in the fusion representations. Specifically, a linear transformation is adopted for three given inputs X_q , X_k and X_v , output with query subspace $Q_i \in R^{d_m \times d_q}$, key subspace $K_i \in R^{d_m \times d_k}$ and value subspace $V_i \in R^{d_m \times d_v}$. The linear transformation is formulated as following:

$$Q_i = X_q W_{qi}, K_i = X_k W_{ki}, V_i = X_v W_{vi} \quad (24)$$

where W_{qi} , W_{ki} , and W_{vi} are learnable parameters. Based on Q_i , K_i , V_i and a scaling factor d_k , the $head_i$ can be calculated as:

$$head_i = \text{softmax}\left(\frac{Q_i (K_i)^T}{\sqrt{d_k}}\right) V_i \quad (25)$$

Then, attention fusion representations of three inputs X_q , X_k and X_v are defined as:

$$Fusion(X_q, X_k, X_v) = \text{concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_n)W_D \quad (26)$$

where *concat* is an operation that concatenates attention features of n heads, and W_D is a learnable parameter used to further fuse concatenating results and reduce dimension.

Based on the global multi-head self-attention mechanism, two semantic representations captured by distance graph and OD graph are mapped into query subspace and key subspace respectively. Semantic representation of distance graph is also used as the value vector. Therefore, the two semantic representations are fused by the global multi-head self-attention mechanism as follows:

$$H_G = Fusion(GCN(A_{dis}, X), GCN(A_{OD}, X), GCN(A_{dis}, X)) \quad (27)$$

where H_G is the final output of global regularity module.

F. Global-Local Information Fusion Module

Having captured the local spatial-temporal relations and global semantic representations, it is critical to properly integrate these heterogeneous dependencies for accurate traffic demand prediction. We propose the global-local information fusion module to fuse the global-local multi-grained features based on a multi-head self-attention mechanism, and one-layer convolution operation is used to conduct multi-step prediction. The aggregated local dependencies H_L and global representations H_G are processed by line transformation, as the input of a multi-head attention mechanism. The specific format is as follows:

$$X_{out} = Fusion(H_L, H_G, H_G) \quad (28)$$

Then, the fused representation X_{out} is fed to two-layer convolution to complete the multi-step prediction. The specific format is as follows:

$$\vec{Y} = \theta_2 * (\theta_1 * X_{out}) \quad (29)$$

where θ_1 and θ_2 are independent convolution operations, and $\vec{Y}$ are the multi-step predicted results.

When in training process, the goal is to optimized the error between predicted results $\vec{Y}$ and ground truth Y . L1 loss is selected as the training objective and the loss function of LPE-STGTN for multi-step prediction can be defined as:

$$L(W_\theta) = \sum_{t=1}^{t=T} |Y_t - \vec{Y}_t| \quad (30)$$

where W_θ denotes all the learnable parameters in the model. The optimization problem can be solved via back-propagation [45] and Adam optimizer [46].

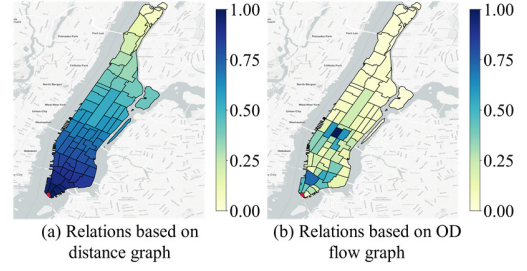


Fig. 6. Relation based on different semantic graphs.

IV. EXPERIMENTS

A. Datasets

Three real-world datasets of New York and Beijing are used to evaluate the performance and applicability of LPE-STGTN. The experiments on two datasets of New York aim at demonstrating the outstanding effectiveness and efficiency of our model both on traditional taxi and ride-hailing datasets, while a taxi dataset of Beijing is selected to validate the wide applicability of our model when applying to another city.

In terms of the two datasets in New York, the first one is a taxi dataset (NYC-Taxi) released by New York Taxi and Limousine Commission (TLC). The dataset collects taxi trip records in New York, including the trip information such as pick-up and drop-off time, origin and destination represented as TLC zone ID. The second one is a for-hire-vehicle dataset (NYC-RH) also released by TLC. The dataset contains the trip information in New York reported by ride-hailing company such as Uber and Lyft, with the origin and destination in the form of TLC zone ID. Compared the methods in zone partition such as grid-based [47] and road-based [48], zone-based [38], [16] is more suited to taxi demand prediction, because the other two methods may break the integrity of urban functional areas crucial for prediction of taxi demand. Therefore, we apply zone-based method to divide the regions of Manhattan as 68 zones, that is, following the existing partition of Manhattan area provided by TLC. Fig.6 shows the relations between randomly selected zone and other zones based on distance graph or OD flow graph. The selected zone is marked with red, and the color of other zones is related with the correlations to selected zone. Fig.6 reflects that in addition to adjacent zones around the selected zone, the remote zones can also have high correlation with the selected zone.

The taxi or ride-hailing orders generating in Manhattan area in 2018 are used as the research data, including a total of 86,442,609 taxi trips and 74,658,668 ride-hailing trips after data filtering. For time granularity, we follow [20] and [26] to aggregate trips at 15-minute intervals. The datasets are split into training, validation and test sets according to 60%, 20% and 20%, respectively, and the length of the input demand sequence is equal to the length of output demand sequence, i.e. $T' = T = 12$ following the previous works [18], [47]. All the data are fed to model after the process of Z-score normalization method as follow:

$$\hat{X} = \frac{X - \text{mean}(X_{train})}{\text{std}(X_{train})} \quad (31)$$

where $\text{mean}(X_{\text{train}})$ and $\text{std}(X_{\text{train}})$ are the mean and standard deviation of training data, and $\hat{X}$ is the true input normalized from original taxi demand sequence X .

B. Baseline Models

To demonstrate the superiority of the proposed model, we compare it with the following representative and state-of-the-art models. Specifically, careful parameter-tuning processes are conducted for all the baselines, and we run their codes based on their recommended configurations or the optimal setting obtained by our experiments. Additionally, early stop strategy with the patience of 30 is adopted for STAEformer [50], while the patience of 20 is set for the other baselines.

LSTM (2014) [50]: The model is a variant of RNN, which is widely used in traffic forecasting. The batch size is 64 and learning rate is 0.001.

STGCN(2018) [33]: The model combines GCN with the one-dimensional convolution for capturing spatial-temporal relations. The batch size is 50 and learning rate is 0.001.

ASTGCN(2019) [34]: The model employs spatial and temporal attention with GCN to extract the dynamic relations. The batch size is 32 and learning rate is 0.001.

Graph WaveNet(2019) [22]: Graph WaveNet (GWNNet) combines fixed graph and self-adaptive graph to capture spatial dependencies and uses TCN to learn temporal relations. The batch size is 64 and learning rate is 0.001.

AGCRN (2020) [21]: The model designs self-adaptive matrix and factorizes the parameters to capture node-specific patterns. The batch size is 64 and learning rate is 0.001.

GMAN (2020) [35]: The model learns spatial-temporal dependencies through stacked spatial and temporal attention layers. The batch size is 16 and learning rate is 0.001.

STSGCN(2020) [19]: The model captures local spatial-temporal correlations through local spatial-temporal graph. The batch size is 32 and learning rate is 0.003.

MVSTT (2022) [28]: The model combines the GCN with transformer to learn different levels of spatial-temporal features. The batch size is 16 and learning rate is 0.0001.

DDSTGCN (2023) [42]: The model extracts dependencies through convolutions on traffic flow graph and its dual hypergraph. The batch size is 64 and learning rate is 0.001.

STAEformer (2023) [49]: A model designs adaptive embedding to enhance the performance and efficiency of vanilla transformer without any graph modeling. The batch size is 64 and learning rate is 0.001. The dimension of feature embedding and adaptive embedding are set as 12 and 60.

PDFormer (2023) [4]: The model learns dynamic spatial dependencies through self-attention, and attempts to explicitly modeling the time delay of spatial information propagation. The batch size is 16 and learning rate is 0.001.

C. Experimental Settings

Our LPE-STGTN model consists of a local spatial-temporal intertwined module, a global spatial-temporal regularity module and a global-local information fusion module. The embedding dimension of the periodicity embedding matrix E_d is set

as 10. The layer of GCRN is 1 and the hidden dimension of it is 64. The size of local window in AFT-local is 4. The number of heads in multi-head attention mechanism in global spatial-temporal regularity module is 4, as well as that in global-local information fusion module. The learning rate is 0.001 and batch size is 64 for all the datasets.

All the models are implemented by the Pytorch framework, and all the experiments are conducted in a Windows environment with the following configuration: 12 vCPU Intel(R) Xeon(R) Platinum 8255C CPU @ 2.50GHz and one NVIDIA GeForce RTX 2080TI GPU. Mean absolute error (MAE), mean absolute percentage error (MAPE), and root mean square error (RMSE) are adopted as the metrics to evaluate the forecasting performances of all deep learning models. Three metrics are formulated as follows:

$$\text{MAE} = \frac{1}{N \times T} \sum_{t=1}^T \sum_{i=1}^N |Y_t^i - \hat{Y}_t^i| \quad (32)$$

$$\text{RMSE} = \sqrt{\frac{1}{N \times T} \sum_{t=1}^T \sum_{i=1}^N (Y_t^i - \hat{Y}_t^i)^2} \quad (33)$$

$$\text{MAPE} = \frac{1}{N \times T} \sum_{t=1}^T \sum_{i=1}^N \frac{|Y_t^i - \hat{Y}_t^i|}{Y_t^i} \quad (34)$$

where T is the number of time steps of an input sequence, N is the number of zones, Y_t^i and $\hat{Y}_t^i$ denote the true and predicted taxi demand in zone i at time step t .

D. Experiment Results

In summary, our LPE-STGTN outperforms all the baselines on most indicators across three datasets. Specifically, compared with PDFormer, LPE-STGTN achieves a balance between effectiveness and efficiency on three datasets, with average training time reduction of 70.66% and average performance improvement of 1.96%. Moreover, the designed spatial-temporal evolving graph generator demonstrates superiority in both effectiveness and efficiency compared to variants equipped with other types of graph generators from existing studies, with average improvements of 2.59% in accuracy and 73.38% in efficiency. Furthermore, several examples are analyzed to provide an intuitive understanding of our model's superior ability in learning local correlations and to enhance its explainability. Detailed analysis is provided as follows.

1) Performance Analysis: We first compare LPE-STGTN with the baseline models on NYC-Taxi and NYC-RH. The average prediction performances of 12 horizons are shown in Table I, and detailed performances in short-term, middle-term and long-term prediction are presented in Table II. The best results are highlighted in bold font, and the second-best results are underlined. In general, it is obvious that our model outperforms all the baselines on most indicators of two datasets.

Specifically, in Table I, the overall performance of LSTM is far inferior to other models due to the lack of ability in extracting spatial correlations from traffic demand data.

TABLE I
TAXI DEMAND PREDICTION COMPARISON OF THE LPE-STGTN MODEL AND OTHER BASELINE MODELS

Dataset	Model	Dynamic dependencies learning	MAE	RMSE	MAPE
NYC-Taxi	LSTM	Time series modeling	7.52	14.16	44.44%
	STGCN	Fixed graph; convolution	6.87	12.62	37.58%
	GWNet	Global adaptive graph and fixed graph; TCN	5.88	10.84	32.76%
	ASTGCN	Fixed graph; spatial-temporal attention; convolution	6.68	12.11	39.86%
	AGCRN	Global adaptive graph; GRU	6.01	11.01	35.59%
	GMAN	Spatial and temporal attention	5.85	10.72	<u>32.39%</u>
	STSGCN	Localized spatial-temporal graph	6.11	11.27	33.82%
	MVSTT	Dynamic graph based on fixed graph and input of each time step	5.88	10.80	33.88%
	DDSTGCN	Sequence-varying graph and global adaptive graph	5.92	10.98	32.86%
	STAEformer	Time-of-day and day-of-week embedding; transformer	6.38	11.21	38.57%
	PDFormer	Time-varying graph based on self-attention with delay features	<u>5.67</u>	10.34	32.42%
	LPE-STGTN	Spatial-temporal evolving graph based on GAT and multi-scale embedding matrices; AFT-local; GRU	5.66	<u>10.40</u>	31.82%
NYC-RH	LSTM	Time series modeling	7.59	12.64	37.30%
	STGCN	Fixed graph; convolution	6.92	11.26	32.98%
	GWNet	Global adaptive graph and fixed graph; TCN	5.90	<u>9.59</u>	28.05%
	ASTGCN	Fixed graph; spatial-temporal attention; convolution	6.57	10.75	30.82%
	AGCRN	Global adaptive graph; GRU	6.33	10.73	28.64%
	GMAN	Spatial and temporal attention	5.96	9.91	28.21%
	STSGCN	Localized spatial-temporal graph	6.34	10.36	29.68%
	MVSTT	Dynamic graph based on fixed graph and input of each time step	6.02	9.85	28.95%
	DDSTGCN	Sequence-varying graph and global adaptive graph	5.95	9.73	<u>28.15%</u>
	STAEformer	Time-of-day and day-of-week embedding; transformer	6.26	9.91	35.13%
	PDFormer	Time-varying graph based on self-attention with delay features	<u>5.85</u>	9.60	28.45%
	LPE-STGTN	Spatial-temporal evolving graph based on GAT and multi-scale embedding matrices; AFT-local; GRU	5.80	9.56	27.18%

TABLE II
PERFORMANCE COMPARISON OF THE LPE-STGTN MODEL AND OTHER BASELINE MODELS ON DIFFERENT TIME HORIZONS

Datasets	Metric	AGCRN	STSGCN	MVSTT	GWNet	GMAN	DDSTGCN	PDFormer	LPE-STGTN	
NYC-Taxi	Horizon 3	MAE	5.55	5.65	5.39	5.41	5.55	5.4	<u>5.31</u>	5.30
		RMSE	9.98	10.26	9.74	9.84	10.05	9.86	9.49	<u>9.60</u>
		MAPE	33.71%	31.91%	31.64%	<u>30.16%</u>	30.81%	30.32%	30.46%	29.99%
	Horizon 6	MAE	5.98	6.12	5.88	5.92	5.83	5.93	5.67	5.67
		RMSE	10.91	11.28	10.75	10.86	10.66	10.97	10.30	<u>10.39</u>
		MAPE	35.39%	33.86%	34.11%	32.65%	<u>31.93%</u>	32.89%	32.15%	31.62%
	Horizon 12	MAE	6.73	6.78	6.67	6.6	6.33	6.7	<u>6.28</u>	6.26
		RMSE	12.49	12.55	12.37	12.27	11.73	12.53	11.62	<u>11.67</u>
		MAPE	38.46%	37.71%	36.77%	<u>36.63%</u>	34.29%	36.31%	35.89%	<u>34.63%</u>
NYC-RH	Horizon 3	MAE	5.68	5.62	5.41	5.28	5.56	<u>5.32</u>	5.36	5.28
		RMSE	9.52	8.90	8.61	<u>8.31</u>	9.10	8.10	8.61	8.44
		MAPE	26.73%	27.02%	26.96%	<u>25.88%</u>	26.96%	26.12%	26.43%	25.66%
	Horizon 6	MAE	6.28	6.31	5.95	5.84	5.92	5.92	5.80	<u>5.81</u>
		RMSE	10.54	10.25	9.65	9.45	9.81	9.62	<u>9.48</u>	9.57
		MAPE	28.28%	28.93%	28.37%	<u>27.65%</u>	28.02%	27.85%	27.98%	26.79%
	Horizon 12	MAE	7.77	7.32	6.96	6.83	<u>6.56</u>	6.85	6.61	6.53
		RMSE	12.39	12.11	11.58	11.26	11.08	11.35	<u>10.99</u>	10.92
		MAPE	32.06%	34.24%	32.22%	31.52%	<u>30.23%</u>	31.82%	31.55%	29.77%

Global adaptive GCN-based models (GWNet and AGCRN) and self-attention-based model (GMAN), which possess superior global perception abilities to learn more information from distant yet similar zones, demonstrate better overall performances than models relying on static graphs (STGCN, ASTGCN). Compared to previous global adaptive GCN and self-attention-based models, MVSTT and DDSTGCN design sequence-varying graph generator and exhibit strong ability in extract dynamic features in taxi demand, obtaining significant

improvements. However, sequence-varying graphs can only learn sequence-specific demand pattern cross multiple time steps, failing to capture the diversified interaction among zones within each time step. The results of LPE-STGTN and PDFormer on two datasets further show the effectiveness of time-varying graph in extracting local spatial-temporal dependencies. The overall performances of our model on two datasets are better than that of PDFormer, with the improvement of 1.19%. Specifically, our model achieves five

TABLE III
THE COMPUTATION COST COMPARISON OF THE LPE-STGTN MODEL
AND OTHER BASELINE MODELS

Models	Computation Time (s/epoch)			
	NYC-Taxi		NYC-RH	
	Training time	Testing time	Training time	Testing time
AGCRN	31.42	3.50	33.79	3.38
STSGCN	68.25	6.11	67.22	6.02
MVSTT	100.54	11.28	113.60	11.41
GWNet	111.90	34.98	118.30	37.10
GMAN	67.25	8.87	67.62	8.87
DDSTGCN	244.21	71.46	241.89	70.16
PDFormer-16	134.27	7.20	132.32	7.26
PDFormer-64	65.54	2.23	72.4	1.87
LPE-STGTN-16	105.86	14.03	107.33	12.61
LPE-STGTN-64	42.47	6.35	42.19	5.57

best and one second best results while PDFormer only obtains one best result and two second best results.

Table II further shows the detailed performances in short-term, middle-term and long-term prediction. It is obvious that our LPE-STGTN model achieves excellent performances in all time scales of two datasets. Specifically, our model achieves five best and four second best indicators on NYC-Taxi, demonstrating overwhelming superiority in both short-term and middle-term predictions, and the outstanding effectiveness of our model in capturing local spatial-temporal dependencies. Moreover, our model can also surpass other baselines in long-term prediction on NYC-RH, where our model achieves all the three best indicators in long-term prediction. Although the performances of PDFormer are relatively close to our model in all the time scales on the NYC-Taxi dataset, it experiences serious performance degradation in all time scales on the NYC-RH dataset, with only one performance indicator achieving the top rank, while our model achieves six best indicators. In conclusion, our LPE-STGTN model exhibits superior performance in all time scales. This is because that the diversified interaction within each time step, inter-information between adjacent time steps and shared patterns in different temporal periods are coupling learned and considered in local spatial-temporal intertwined module.

2) *Efficiency Analysis*: We further analyze the computational efficiency of all the models with the optimal configuration as mentioned in above baseline setting, in order to provide a comprehensive evaluation of all the models, as illustrated in Table III. On the one hand, as mentioned in above experiments, PDFormer achieves good performances but comes with a significant increase in computational overhead due to the inherent limitation of transformers in efficiency, which requires the longest computation time except DDSTGCN. On the other hand, while AGCRN is the most efficient model, it lacks in capturing dynamic dependencies necessary for achieving excellent performance. In contrast, LPE-STGTN strikes a balance between effectiveness and efficiency, which not only outperforms all the baselines on most indicators across two datasets but also significantly reduces computation time.

Besides comparing the efficiency between PDFormer and LPE-STGTN with their optimal configurations, we also test the computation time of PDFormer with the batch size of 64 and

TABLE IV
ABLATION EXPERIMENT ON NYC-TAXI AND NYC-RIDE-HAILING
DATASETS

Model	Dataset					
	NYC-Taxi			NYC-RH		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
Global-Full	6.05	11.21	33.80	6.17	10.26	28.07
Local-Full	5.8	10.69	32.47	5.89	9.69	28.63
LPE-STGTN	5.66	10.40	31.82	5.80	9.56	27.18

LPE-STGTN with the batch size of 16 for a comprehensive evaluation, aiming to exclude the effect of different batch size on computation time. Though the computational efficiency of LPE-STGTN is decreased when trained with batch size of 16, it is more efficient than PDFormer with the same batch size. Similarly, our model still maintains the efficiency improvement of 38.46% compared with PDFormer trained with batch size of 64. It easy to draw that no matter what the batch size is, our model can always achieve more efficient performance than PDFormer. Moreover, while PDFormer exhibits higher testing efficiency than LPE-STGTN with the same batch size, the testing time of our model remains acceptable for deployment in real-world scenarios, due to its predictable three-hour time span. In conclusion, in comparison to PDFormer with optimal setting, the training time of LPE-STGTN with optimal configuration on the two datasets is reduced by 68.37% and 68.15%, respectively.

3) *Ablation Experiment*: We design following two variants of LPE-STGTN to explore the effectiveness of each module in our model:

Global-Full: LPE-STGTN only contains global module and information fusion module, without local module.

Local-Full: LPE-STGTN only contains local module and information fusion module, without global module.

All variants have the same settings as LPE-STGTN, and the performances on two datasets are shown in Table IV. It is easy to draw the conclusion that the complete LPE-STGTN model can achieve the best performance compared with the two variants removing certain components. Specifically, both Global-Full and Local-Full suffer performance degradation, indicating the necessity of coupling consideration of spatial-temporal dependencies from global and local perspectives. Moreover, compared with Global-Full focusing on the extracting global dependencies, the performance of Local-Full is closer to LPE-STGTN, and already better than many excellent baselines such as DDSTGCN and MVST. It not only indicates that the key in prediction of taxi demand is effectively mining intricate local spatial-temporal relations, but also validates the advanced ability of designed local module in learning local dynamic dependencies.

4) *Analysis of Dynamic Graph Generator*: In this subsection, we evaluate the effectiveness and efficiency of the designed spatial-temporal evolving graph generator by replacing it with static graph or other types of graph generators

TABLE V

THE PERFORMANCE COMPARISON BETWEEN LOCAL-FULL AND THE VARIANTS REPLACING THE SPATIAL-TEMPORAL EVOLVING GRAPH GENERATOR WITH OTHER DYNAMIC GRAPH GENERATORS

	Model	Horizon 3			Horizon 6			Average			Computation (s/epoch)
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	Training time
NYC-TAXI	Local-Static	5.90	10.66	34.47%	6.60	12.05	37.96%	6.66	12.19	38.27%	11.87
	Local-Linear	5.51	9.91	31.74%	6.00	11.02	34.36%	6.01	11.06	34.39%	163.65
	Local-SA	5.56	10.11	31.85%	6.15	11.27	34.76%	6.17	11.36	35.00%	162.60
	Local-Hidden	<u>5.47</u>	<u>9.90</u>	<u>31.49%</u>	<u>5.98</u>	<u>11.04</u>	<u>33.48%</u>	<u>5.95</u>	<u>11.00</u>	<u>33.34%</u>	158.85
	Local-Full-II	5.41	9.77	30.54%	5.92	10.90	32.99%	5.91	10.90	32.73%	30.07
	Compared with Local-Hidden	1.10%	1.31%	0.30%	1.00%	1.23%	1.46%	0.67%	0.91%	1.83%	81.07%
NYC-RH	Local-Static	5.88	9.42	29.00	6.68	10.85	32.01%	6.77	11.08	32.34%	39.30
	Local-Linear	5.49	8.72	26.69	6.14	10.01	28.80%	6.16	10.06	<u>29.12%</u>	141.54
	Local-SA	5.51	8.77	26.81%	6.17	10.07	<u>29.26%</u>	6.24	10.25	29.43%	139.61
	Local-Hidden	<u>5.45</u>	<u>8.66</u>	<u>26.53%</u>	<u>6.03</u>	<u>9.75</u>	29.37%	<u>6.14</u>	<u>10.03</u>	29.59%	211.12
	Local-Full-II	5.36	8.47	26.33%	5.87	9.49	28.19%	5.91	9.59	28.40%	72.45
	Compared with Local-Hidden	1.65%	2.19%	0.75%	2.65%	2.67%	4.02%	3.75%	4.39%	4.02%	65.68%

from existing studies. To validate the spatial-temporal dynamic learning ability of each type of graph generator more strictly, day-of-week index and time-of-day index are not considered in input, and only the taxi demand is fed into models. Local-Full-II is a variant of Local-Full, where day-of-week index and time-of-day index are not contained in input. All the variants are constructed based on Local-Full-II.

Local-Static: The variant replaces the designed spatial-temporal evolving graph generator with static graph based on distance.

Local-Linear: The variant uses linear transformation generator to generate time-varying graph according to [24].

Local-SA: The variant applies self-attention mechanism to generating time-varying graph according to [4].

Local-Hidden: The variant generates time-varying graph based on current input and hidden state of former time step. The graph generator is constructed based on self-attention mechanism, distance-based GCN, hidden states and learnable parameter matrix according to [26].

The performances of all the variants on two datasets of New York are shown in Table V. The performance of Local-Static is far inferior to that of other variants, demonstrating that static graph is hard to model the dynamic interaction among zones along time axis. Additionally, we observe that Local-SA yields the worst results on both datasets among the variants based on the time-varying graph generator. This indicates that relying solely on the self-attention mechanism fails to effectively generate high-quality dynamic graphs. One potential reason is that the self-attention mechanism tends to focus on learning global components rather than extracting local features required for generating dynamic graphs. In contrast, Local-Hidden, which achieves most of the second-best results on both two datasets, incorporates local information into the self-attention mechanism by combining its output with hidden states of previous time step. Although Local-Linear,

Local-SA and Local-Hidden show significant performance improvement compared with Local-Static, all of them are stuck in the dilemma of inefficiency, requiring more than tenfold computation time. Local-Full-II, equipped with the designed spatial-temporal evolving graph generator, exhibits superiority in effectiveness and efficiency, with the average improvements of 2.59% in accuracy and 73.38% in efficiency on two datasets, compared with Local-Hidden. Moreover, Local-Full-II achieves the best results both short-term, middle-term and even the overall prediction cross 12 horizons. The experiment strongly proves the advanced performance of the designed spatial-temporal evolving graph generator, which can effectively and efficiently extract local spatial-temporal dependencies in taxi demand.

E. Analysis of Local Performance and Learned Correlations

The above experiments demonstrate the effectiveness and efficiency of our model in learning local dynamic features from the perspective of overall performance. In this subsection, several examples are analyzed to provide an intuitive understanding of our model's superior ability in learning local correlations and to enhance its explainability. Furthermore, the local performance of our model is further evaluated to validate its dynamic modeling ability. Local-Hidden, the variant achieving the second-best results, is selected to compare the local performance with that of Local-Full-II. The following analysis is conducted on the test set of NYC-RH.

1) *Learned Interaction Among Zones Analysis:* Fig. 7 illustrates the comparison of learned correlations between Local-Full-II and Local-Hidden. Specifically, Fig. 7 (a) displays the interaction between zone 16 and other zones during peak morning hours as learned by Local-Full-II, while Fig. 7 (b) shows the destination distribution of trips originating from zone 16 during peak morning hours. The intensity of interaction/travel is indicated by the darkness of the color, with

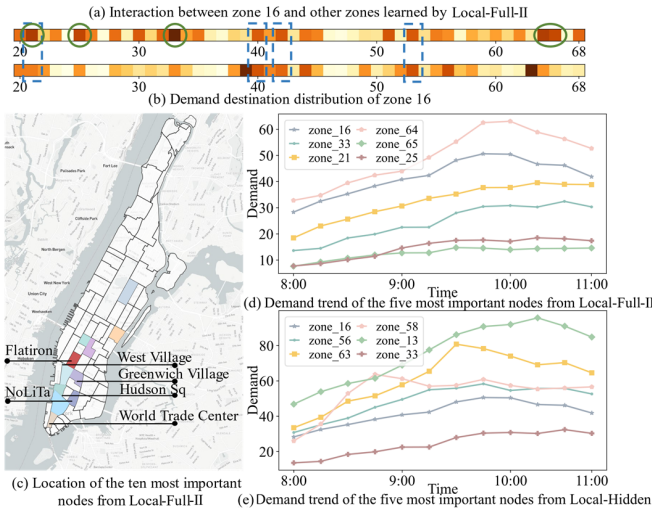


Fig. 7. The comparison of learned correlations between Local-Full-II and Local-Hidden during peak morning hours.

darker shades representing higher intensity. Clearly, we can observe that the four zones in the blue rectangle are the hot destinations of zone 16 in selected period, meanwhile, Local-Full-II learns these travel relationships, where the zones in the blue rectangle account for four of the top ten areas having the strongest interactions with zone 16.

The zones in the green circle are the five most important zones about zone 16 learned by Local-Full-II. Fig. 7 (d) and Fig. 7 (e) illustrate the demand trends of the five most important zones learned by Local-Full-II and Local-Hidden. Intuitively, the similarity in demand trend between the five most important zones and zone 16 in Fig. 7 (d) is notably higher than that in Fig. 7 (e). Furthermore, Pearson correlation coefficient (PCC) is used as the metric to quantitatively evaluate the trend similarity. The average PCC between the demand trends of five most important zones learned by Local-Full-II and that of zone 16 achieves 93.60%, while the similarity of demand trends learned by Local-Hidden is 90.82%.

Fig. 7 (c) illustrates the location of the ten most important zones learned by Local-Full-II, with zone 16 highlighted in red and the remaining zones marked in various colors. The names of zone 16 and the five most crucial zones are also labeled. Specifically, zone 16 is Flatiron district, primarily composed of commercial and office spaces, as well as a few residential spaces. Likewise, the other five most important zones, such as World Trade Center and North of Little Italy (Nolita), are also characterized by business, entertainment, and residential functions. Hence, these zones share similar characteristics and demand patterns with zone 16, represented as the strong correlation in dynamic graph.

It's easy to draw the conclusion that Local-Full-II can explicit extract the mobility patterns, demand patterns and regional functions during the morning peak hours in taxi demand, demonstrating the superiority in dynamically modeling the local spatial-temporal dependencies.

2) *Local Performance Analysis:* Fig. 8 shows the comparison of local performance between Local-Full-II and Local-Hidden. In particular, Fig. 8 (a) and Fig. 8 (c) display two approaches' average performances during evening peak hours

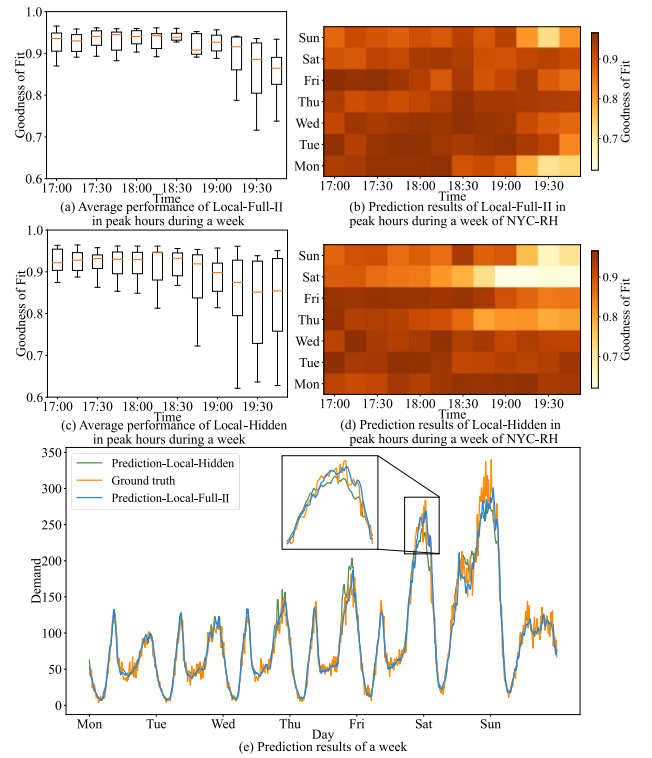


Fig. 8. The comparison of local performance between Local-Full-II and Local-Hidden.

of one week, where goodness of fit is selected as the metric to measure the difference between the predicted and the actual value. Clearly, the prediction performance of Local-Full-II outperforms that of Local-Hidden on most of the time step whether the maximum, minimum or the mean of goodness of fit. Fig. 8 (b), and Fig. 8 (d) further illustrates the detailed performance differences of evening peak hours in each day in a week. It is obvious that Local-Hidden suffers significant performance degradation when predicting the demand of weekends, while Local-Full-II can achieve superior performance on weekdays and weekends throughout. It is a strong proof that the designed multi-scale embedding matrices can effectively learn the diversified patterns of every day in a week.

Fig. 8 (e) compares the prediction results with ground truth to give an intuitive impression of the ability difference of two approaches in extracting local dynamic dependencies. Apparently, both the two approaches can capture the trends and fluctuations well in the first three days. However, the performance of Local-Hidden becomes worse as the weekends approach. Similarly, though faced with slight decrease in performance, Local-Full-II can still effectively extract the demand trend and local fluctuations, which is also served as a solid prove to demonstrate the excellent ability of the designed spatial-temporal graph generator in dynamic modeling.

F. Applicability Analysis on a Dataset of Beijing

A dataset of taxi demand in Beijing city (BJ-Taxi), China, is selected as the supplementary dataset to test the generalization and transferring ability of our model when applying to a different city. The study area of this dataset is situated in the downtown of Beijing, starting from 116.05°E

TABLE VI
DETAILED INTRODUCTION OF BJ-TAXI

Dataset	BJ-Taxi
Number of orders	14,968,765
Number of zones	36
Time span	31 days
Length of time interval	15 min

TABLE VII
TAXI DEMAND PREDICTION COMPARISON OF THE LPE-STGTN AND
PDFORMER ON BJ-TAXI

Time step	PDFormer			LPE-STGTN		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
1	10.62	35.77	32.66	10.06	34.36	31.00%
3	11.69	43.27	36.50	11.10	42.85	32.01%
6	12.00	50.41	38.03	12.93	50.55	37.64%
9	14.76	55.84	41.17	14.37	56.90	38.06%
12	16.61	62.49	48.63	16.21	61.28	40.47%
Average	13.50	51.03	38.62	13.11	51.17	36.46%
Computation (s/epoch)	10.86			2.66		
Performance improvement	2.73%					
Efficiency improvement	75.51%					

to 116.80°E in longitude, and from 39.65°N to 40.25°N in latitude. Additionally, the study area of Beijing is supposed to conduct area division manually so as to correlate the OD with zone ID. Referring to [51], the study area is equally partitioned into 6×6 grids, where each zone is about a square with sides 10 kilometers. The data processing and division toward BJ-Taxi dataset is totally same as that of datasets in New York. Detailed introduction of BJ-Taxi is shown in Table VI.

PDFormer, one of the most competitive baselines, is also selected for comparison with LPE-STGTN when applied to BJ-Taxi. The results are shown in Table VII. It is evident that LPE-STGTN exhibits overwhelming superiority in short-term prediction, as demonstrated by the all the indicators at time step 1 and 3, owing to its robust ability to extract local spatial-temporal relations. When conducting middle-term and long-term prediction, our model still maintains significant advantages in MAE and MAPE, while PDFormer achieves slightly better performance in RMSE. This suggests that the designed local spatial-temporal module presents noteworthy ability in modeling short-, middle- and long-term dependencies as well, even when compared with transformer-based model with outstanding global perception. It is worth noting that in comparison with that of PDFormer, LPE-STGTN significantly achieves a balance between effectiveness and efficiency on BJ-Taxi, with average training time reduction of 75.51% and average performance improvement of 2.73%.

In conclusion, LPE-STGTN can achieve remarkable performances while maintaining high efficiency on datasets from different cities, demonstrating its broad applicability and superior practicality.

V. CONCLUSION

In this research, a novel LPE-STGTN model is proposed to improve the effectiveness and efficiency in extracting intricate

local dynamics in taxi demand. In the local spatial-temporal intertwined module, a spatial-temporal evolving graph generator is designed for describing the dynamic interaction among zones along time axis, where the multi-scale embedding matrices are designed to efficiently absorb shared and diversified inter-node relations cross different temporal periodicity, meanwhile GAT is used to self-learn the specific spatial interaction among nodes within each time step. Moreover, AFT-local is introduced to effectively and efficiently learn the information passing between adjacent time steps. Global dependencies are also learned by dual semantic GCN and global multi-head self-attention mechanism.

We evaluate the proposed LPE-STGTN on three real-world taxi demand datasets in New York and Beijing. The experimental results demonstrate the following: (1) LPE-STGTN outperforms all the baselines on most indicators across three datasets, indicating its advanced ability to learn dynamic spatial-temporal dependencies. (2) Compared to PDFormer, LPE-STGTN can achieve a balance between effectiveness and efficiency on three datasets, with average training time reduction of 70.66% and average performance improvement of 1.96%. (3) Analysis of the dynamic graph generator indicates that the designed spatial-temporal evolving graph generator demonstrates superiority in both effectiveness and efficiency, compared to variants equipped with other types of graph generators from existing studies, with average improvements of 2.59% in accuracy and 73.38% in efficiency. (4) The designed local module is highly interpretable, explicitly extracting mobility patterns, demand patterns, regional functions, as well as learning demand trends and local fluctuations. The proposed LPE-STGTN can serve as a technological foundation for large-scale traffic prediction in future traffic systems with mixed levels of vehicle autonomy.

The experimental results confirm that effectively and efficiently learning local spatial-temporal dependencies plays a crucial role in enhancing the accuracy of traffic forecasting. There is still room for further improvement due to the diversified factors influencing the forecasting accuracy. External conditions such as weather, accidents and special events can influence travel demand. Additionally, the quality of data almost determines the upper limit forecasting performance, thus excellent data pre-processing and data augmentation methods may be the key to improving performance. Moreover, a scientific method to determine the appropriate granularity of time and spatial may also be a promising way to further improving the forecasting performance in future work.

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